

# Section 10

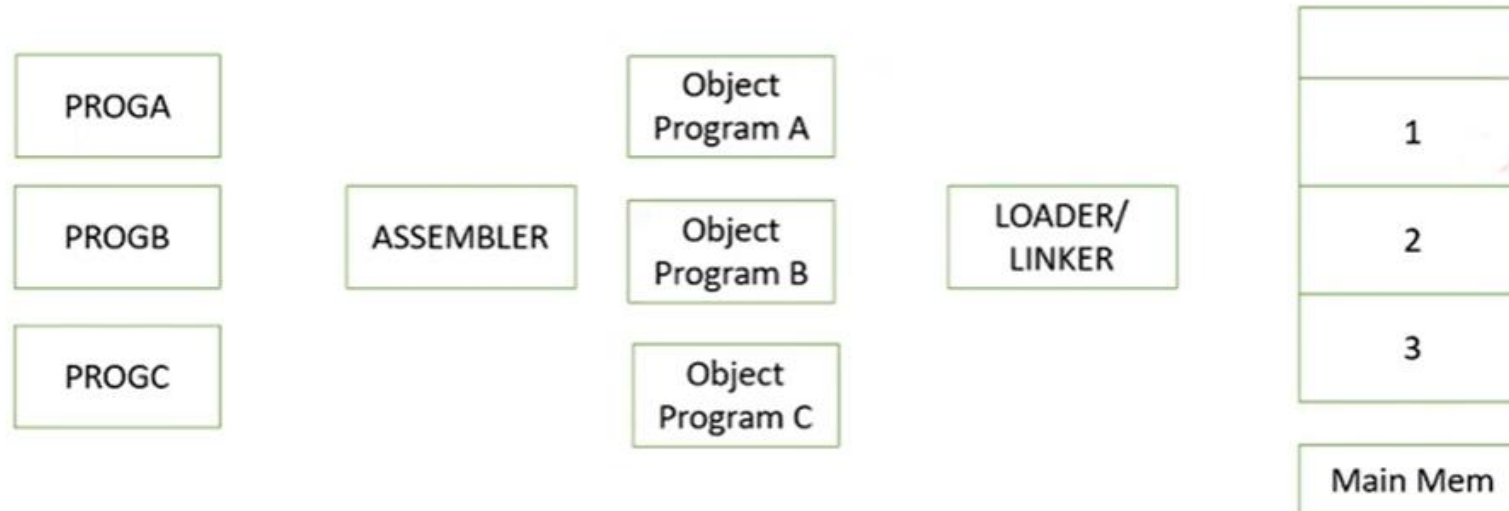
Systems Programming ECE3502

# Loaders and Linkers

**Loading:** loading an object program into memory for execution.

**Relocation:** modify the object program so that it can be loaded at an address from the location originally specified.

**Linking:** combines two or more separate object programs and supplies the information needed to allow references between them.



# Absolute loader

**Q1: Apply the absolute loader for the following H.T.E record:**

H COPY 001000 00107A

T 001000 1E 141033 482039 001036 281030 301015 482061 3C1003 00102A 0C1039 00102D

T 01001E 15 0C1036 482061 081033 4C0000 454F46 000003 000000

T 002039 1E 041030 001030 E0205D 30203F D8205D 281030 302057 549039 2C205E 38203F

T 002057 1C 101036 4C0000 F1 001000 041030 E02079 302064 509039 DC2079 2C1036

T 002073 07 382064 4C0000 05

E 001000

# Absolute loader

| Memory Address | Contents |          |          |          |
|----------------|----------|----------|----------|----------|
| <u>0000</u>    | XXXXXXXX | XXXXXXXX | XXXXXXXX | XXXXXXXX |
| <u>0010</u>    | XXXXXXXX | XXXXXXXX | XXXXXXXX | XXXXXXXX |
| <u>0020</u>    | XXXXXXXX | XXXXXXXX | XXXXXXXX | XXXXXXXX |
| <u>0030</u>    | XXXXXXXX | XXXXXXXX | XXXXXXXX | XXXXXXXX |
| . -            | .        | .        | .        | .        |
| .              | .        | .        | .        | .        |
| .              | .        | .        | .        | .        |
| <u>1000</u>    | 14103348 | 20390010 | 36281030 | 30101548 |
| <u>1010</u>    | 20613C10 | 0300102A | 0C103900 | 102D0C10 |
| <u>1020</u>    | 36482061 | 0810334C | 0000454F | 46000003 |
| <u>1030</u>    | 000000xx | XXXXXXXX | XXXXXXXX | XXXXXXXX |
| . -            | .        | .        | .        | .        |
| .              | .        | .        | .        | .        |
| .              | .        | .        | .        | .        |
| <u>2030</u>    | XXXXXXXX | XXXXXXXX | xx041030 | 001030E0 |
| <u>2040</u>    | 205D3020 | 3FD8205D | 28103030 | 20575490 |
| <u>2050</u>    | 392C205E | 38203F10 | 10364C00 | 00F10010 |
| <u>2060</u>    | 00041030 | E0207930 | 20645090 | 39DC2079 |
| <u>2070</u>    | 2C103638 | 20644C00 | 0005XXXX | XXXXXXXX |
| <u>2080</u>    |          |          |          |          |

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**H** COPY 001000 00107A

**T** 001000 1E 141033 482039 001036  
281030 301015 482061 3C1003  
00102A 0C1039 00102D

**T** 01001E 15 0C1036 482061 081033  
4C0000 454F46 000003 000000

**T** 002039 1E 041030 001030 E0205D  
30203F D8205D 281030 302057  
549039 2C205E 38203F

**T** 002057 1C 101036 4C0000 F1  
001000 041030 E02079 302064  
509039 DC2079 2C1036

**T** 002073 07 382064 4C0000 05

**E** 001000

Q2: Consider the three following programs. Suppose they are linked in the sequenced of **PROG2+PROG3+PROG1**

Show the memory content for all programs after the operation of Absolute loader

```
=====
H.XXPROG1.0000000.00014E
D.XALPHA.00001F.XXXMAX.00014B
R.XXXINF.XXZERO
T.000000.1D.6D000375012C05000003A0132B213C2B30
00160F21359041A0153B2FEA
T.00014B.03.000000
M.000010.05+PROG1
M.00014B.06+INF
M.00014B.06ZERO
E.000000
=====
H.XXPROG2.0000000.000026
D.XXXMIN.000023.XXZERO.00001D
R.XALPHA.XXXMAX
T.000000.1E.6F201D75012C050000039000002BA01337
20030F200D9041A0153B2FEC00
```

```
T.00001E.09.0000000000003000000
M.00000A.05+ALPHA
M.000023.06+MAX
E.000000
=====
H.XXPROG3.0000000.000023
D.XXXINF.000020
R.02XXXMAX.03XXXMIN
T.000000.18.031000006F1000009C400C10001898406F20089
404FB2005
T.000020.03.008000
M.000001.05+02
M.000005.05+03
M.00000B.05+01
E.000000
```

Step 1: Load the T- record of PROG2 in the memory

H.XXPROG2.0000000.000026

D.XXXMIN.000023.XXZERO.00001D

R.XALPHA.XXXMAX

T.000000.1E.6F201D75012C050000039000002BA0133720030F200D9041A0153B2FEC00

T.00001E.09.000000000003000000

M.00000A.05+ALPHA

M.000023.06+MAX

E.000000

[illegible]

Step 2: Load the T- record of PROG3 in the memory

H.XXPROG3.0000000.000023

D.XXXINF.000020

R.02XXXMAX.03XXXMIN T.000000.18.031000006F1000009C400C10001898406F20089404FB2005

T.000020.03.008000

M.000001.05+02

M.000005.05+03

M.00000B.05+01

E.000000

[illegible]

## Step 3: Load the T- record of PROG1 in the memory

H.XXPROG1.0000000.00014E

D.XALPHA.00001F.XXXMAX.00014B

R.XXXINF.XXZERO

T.000000.1D.6D000375012C05000003A0132B213C2B3000160F21359041A0153B2FEA

T.00014B.03.000000

M.000010.05+PROG1

M.00014B.06+INF

M.00014B.06ZERO

E.000000

| ADDRESS | Content |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---------|---------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
|         | 0       | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | A  | B  | C  | D  | E  | F  |
| 0000    | 6D      | 00 | 03 | 75 | 01 | 2C | 05 | 00 | 00 | 03 | A0 | 13 | 2B | 21 | 3C | 2B |
| 0010    | 30      | 00 | 16 | 0F | 21 | 35 | 90 | 41 | A0 | 15 | 3B | 2F | EA | 00 | 00 | 00 |
| 0020    | 00      | 80 | 00 | 03 | 00 | 00 | 00 |    |    |    |    |    |    |    |    |    |
| .       |         |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| .       |         |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| 0140    |         |    |    |    |    |    |    |    |    |    |    | 00 | 00 | 00 |    |    |



# Loader and Linker

| Loc  | Source statement |                                                                             | Object code |                                                                                                                                                                                                                                                                        |
|------|------------------|-----------------------------------------------------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0000 | PROGA            | START .0<br>EXTDEF LISTA, ENDA<br>EXTREF LISTB, ENDB, LISTC, ENDC<br>.<br>. |             | HPROGA 0000000000063<br>DLISTA 000040^ ENDA 000054<br>RLISTB ^ ENDB ^ LISTC ^ ENDC<br>:<br>T0000200A03201D77100004050014<br>:                                                                                                                                          |
| 0020 | REF1             | LDA LISTA                                                                   | 03201D      | :                                                                                                                                                                                                                                                                      |
| 0023 | REF2             | +LDT LISTB+4                                                                | 77100004    |                                                                                                                                                                                                                                                                        |
| 0027 | REF3             | LDX #ENDA-LISTA                                                             | 050014      | T0000540F000014^ FFFFF6000003F000014^ FFFFC0<br>M00002405+LISTB<br>M00005406+LISTC<br>M00005706+ENDC<br>M00005706-LISTC<br>M00005A06+ENDC<br>M00005A06-LISTC<br>M00005A06+PROGA<br>M00005D06-ENDB<br>M00005D06+LISTB<br>M00006006+LISTB<br>M00006006-PROGA<br>E000020^ |
| 0040 | LISTA            | EQU *                                                                       |             |                                                                                                                                                                                                                                                                        |
| 0054 | ENDA             | EQU *                                                                       |             |                                                                                                                                                                                                                                                                        |
| 0054 | REF4             | WORD ENDA-LISTA+LISTC                                                       | 000014      |                                                                                                                                                                                                                                                                        |
| 0057 | REF5             | WORD ENDC-LISTC-10                                                          | FFFFFF6     |                                                                                                                                                                                                                                                                        |
| 005A | REF6             | WORD ENDC-LISTC+LISTA-1                                                     | 00003F      |                                                                                                                                                                                                                                                                        |
| 005D | REF7             | WORD ENDA-LISTA- (ENDB-LISTB)                                               | 000014      |                                                                                                                                                                                                                                                                        |
| 0060 | REF8             | WORD LISTB-LISTA                                                            | FFFC0       |                                                                                                                                                                                                                                                                        |
|      |                  | END REF1                                                                    |             |                                                                                                                                                                                                                                                                        |

# Loader and Linker

| Loc  | Source statement |        |                          | Object code |                                     |
|------|------------------|--------|--------------------------|-------------|-------------------------------------|
| 0000 | PROGB            | START  | 0                        |             | HPROGB 000000000007F                |
|      |                  | EXTDEF | LISTB, ENDB              |             | DLISTB 000060ENDB 000070            |
|      |                  | EXTREF | LISTA, ENDA, LISTC, ENDC |             | RLISTA ENDA LISTC ENDC              |
|      |                  | .      |                          |             | .                                   |
|      |                  | .      |                          |             | .                                   |
|      |                  | .      |                          |             | T0000360B0310000077202705100000     |
|      |                  | .      |                          |             | .                                   |
| 0036 | REF1             | +LDA   | LISTA                    | 03100000    | T0000700F000000FFFFF6FFFFFFF0000060 |
| 003A | REF2             | LDT    | LISTB+4                  | 772027      | M00003705+LISTA                     |
| 003D | REF3             | +LDX   | #ENDA-LISTA              | 05100000    | M00003E05+ENDA                      |
|      |                  | .      |                          |             | M00003E05-LISTA                     |
|      |                  | .      |                          |             | M00007006+ENDA                      |
|      |                  | .      |                          |             | M00007006-LISTA                     |
|      |                  | .      |                          |             | M00007006+LISTC                     |
| 0060 | LISTB            | EQU    | *                        |             | M00007306+ENDC                      |
|      |                  | .      |                          |             | M00007306-LISTC                     |
|      |                  | .      |                          |             | M00007606+ENDC                      |
| 0070 | ENDB             | EQU    | *                        |             | M00007606-LISTC                     |
| 0070 | REF4             | WORD   | ENDA-LISTA+LISTC         | 000000      | M00007606+LISTA                     |
| 0073 | REF5             | WORD   | ENDC-LISTC-10            | FFFFFF6     | M00007906+ENDA                      |
| 0076 | REF6             | WORD   | ENDC-LISTC+LISTA-1       | FFFFFFF     | M00007906-LISTA                     |
| 0079 | REF7             | WORD   | ENDA-LISTA- (ENDB-LISTB) | FFFFFF0     | M00007C06+PROGB                     |
| 007C | REF8             | WORD   | LISTB-LISTA              | 000060      | M00007C06-LISTA                     |
|      |                  | END    |                          |             | E                                   |

# Loader and Linker

| Loc  |       | Source statement                | Object code |                                         |
|------|-------|---------------------------------|-------------|-----------------------------------------|
| 0000 | PROGC | START 0                         |             | HPRGCG 000000000051                     |
|      |       | EXTDEF LISTC, ENDC              |             | DLISTC 000030ENDC 000042                |
|      |       | EXTREF LISTA, ENDA, LISTB, ENDB |             | RLISTA ENDA LISTB ENDB                  |
|      |       | .                               |             | :                                       |
|      |       | .                               |             | :                                       |
|      |       | .                               |             | :                                       |
|      |       | .                               |             | :                                       |
|      |       | .                               |             | :                                       |
| 0018 | REF1  | +LDA LISTA                      | 03100000    | T0000180C031000007710000405100000       |
| 001C | REF2  | +LDT LISTB+4                    | 77100004    | :                                       |
| 0020 | REF3  | +LDX #ENDA-LISTA                | 05100000    | T0000420F000030000008000011000000000000 |
|      |       | .                               |             | M00001905+LISTA                         |
|      |       | .                               |             | M00001D05+LISTB                         |
|      |       | .                               |             | M00002105+ENDA                          |
|      |       | .                               |             | M00002105-LISTA                         |
|      |       | .                               |             | M00004206+ENDA                          |
|      |       | .                               |             | M00004206-LISTA                         |
| 0030 | LISTC | EQU *                           |             | M00004206+PROGC                         |
|      |       | .                               |             | M00004806+LISTA                         |
|      |       | .                               |             | M00004B06+ENDA                          |
| 0042 | ENDC  | EQU *                           |             | M00004B06-LISTA                         |
| 0042 | REF4  | WORD ENDA-LISTA+LISTC           | 000030      | M00004B06-ENDB                          |
| 0045 | REF5  | WORD ENDC-LISTC-10              | 000008      | M00004B06+LISTB                         |
| 0048 | REF6  | WORD ENDC-LISTC+LISTA-1         | 000011      | M00004E06+LISTB                         |
| 004B | REF7  | WORD ENDA-LISTA- (ENDB-LISTB)   | 000000      | M00004E06-LISTA                         |
| 004E | REF8  | WORD LISTB-LISTA                | 000000      | E                                       |
|      |       | END                             |             |                                         |

# Relocatable Loaders (Linking Loader)

LOAD PROG A + PROG B +PROG C (Start from Address 4000)

| Control Section | Symbol name | Address      | Length |
|-----------------|-------------|--------------|--------|
| <b>PROGA</b>    |             | 4000         | 0063   |
|                 | LISTA       | 4000+40=4040 |        |
|                 | ENDA        | 4000+54=4054 |        |
| <b>PROGB</b>    |             | 4063         | 007F   |
|                 | LISTB       | 4063+60=40C3 |        |
|                 | ENDB        | 4063+70=40D3 |        |
| <b>PROGC</b>    |             | 40E2         | 0051   |
|                 | LISTC       | 40E2+30=4112 |        |
|                 | ENDC        | 40E2+42=4124 |        |

# Relocatable Loaders (Linking Loader)

## Loading PROGA:

| ADDRESS | Content |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---------|---------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
|         | 0       | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | A  | B  | C  | D  | E  | F  |
| 4000    |         |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| 4010    |         |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| 4020    | 03      | 20 | 1D | 77 | 10 | 00 | 04 | 05 | 00 | 14 |    |    |    |    |    |    |
| .       |         |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| 4050    |         |    |    |    | 00 | 00 | 14 | FF | FF | F6 | 00 | 00 | 3F | 00 | 00 | 14 |
| 4060    | FF      | FF | C0 |    |    |    |    |    |    |    |    |    |    |    |    |    |

```
HPROGA 0000000000063
DLISTA 000040END 000054
RLISTB ENDB LISTC ENDC
:
T0000200A03201D77100004050014
:
T0000540F000014FFFFFF600003F000014FFFFC0
M00002405+LISTB
M00005406+LISTC
M00005706+ENDC
M00005706-LISTC
M00005A06+ENDC
M00005A06-LISTC
M00005A06+PROGA
M00005D06-ENDB
M00005D06+LISTB
M00006006+LISTB
M00006006-PROGA
E000020
```

# Relocatable Loaders (Linking Loader)

## Updating the memory for PROGA:

| ADDRESS | Content |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---------|---------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
|         | 0       | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | A  | B  | C  | D  | E  | F  |
| 4000    |         |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| 4010    |         |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| 4020    | 03      | 20 | 1D | 77 | 10 | 40 | C7 | 05 | 00 | 14 |    |    |    |    |    |    |
| .       |         |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| 4050    |         |    |    |    | 00 | 41 | 26 | 00 | 00 | 08 | 00 | 40 | 51 | 00 | 00 | 04 |
| 4060    | 00      | 00 | 83 |    |    |    |    |    |    |    |    |    |    |    |    |    |

### Modification of PROGA:

| Address      | Operation              | Value   |
|--------------|------------------------|---------|
| 4000+24=4024 | 00004+40C3=            | 040C7   |
| 4000+54=4054 | 000014+4112=           | 004126  |
| 4000+57=4057 | FFFFFF6+4124-4112=     | 0000008 |
| 4000+5A=405A | 00003F+4124-4112+4000= | 004051  |
| 4000+5D=405D | 000014-40D3+40C3=      | 000004  |
| 4000+60=4060 | FFFFC0+40C3-4000       | 000083  |



# Loader and Linker

## Loading PROGB:

| ADDRESS | Content |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---------|---------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
|         | 0       | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | A  | B  | C  | D  | E  | F  |
| 4080    |         |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| 4090    |         |    |    |    |    |    |    |    |    | 03 | 10 | 00 | 00 | 77 | 20 | 27 |
| 40A0    | 05      | 10 | 00 | 00 |    |    |    |    |    |    |    |    |    |    |    |    |
| .       |         |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| 40D0    |         |    |    | 00 | 00 | 00 | FF | FF | F6 | FF | FF | FF | FF | FF | F0 | 00 |
| 40E0    | 00      | 60 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |

```

HPROGB 0000000000007F
DLISTB 000060ENDB 000070
RLISTA ENDA LISTC ENDC
:
:
T0000360B0310000077202705100000
:
T0000700F000000FFFFF6FFFFFFF0000060
M00003705+LISTA
M00003E05+ENDA
M00003E05-LISTA
M00007006+ENDA
M00007006-LISTA
M00007006+LISTC
M00007306+ENDC
M00007306-LISTC
M00007606+ENDC
M00007606-LISTC
M00007606+LISTA
M00007906+ENDA
M00007906-LISTA
M00007C06+PROGB
M00007C06-LISTA
E

```

# Loader and Linker

## Updating the memory for PROGB:

| ADDRESS | Content |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---------|---------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
|         | 0       | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | A  | B  | C  | D  | E  | F  |
| 4080    |         |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| 4090    |         |    |    |    |    |    |    |    |    | 03 | 10 | 40 | 40 | 77 | 20 | 27 |
| 40A0    | 05      | 10 | 00 | 14 |    |    |    |    |    |    |    |    |    |    |    |    |
| .       |         |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| 40D0    |         |    |    | 00 | 41 | 26 | 00 | 00 | 08 | 00 | 40 | 51 | 00 | 00 | 04 | 00 |
| 40E0    | 00      | 83 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |

### Modification of PROGB:

| Address      | Operation               | Value  |
|--------------|-------------------------|--------|
| 4063+37=409A | 00000+4040=             | 04040  |
| 4063+3E=40A1 | 00000+4054-4040=        | 00014  |
| 4063+70=40D3 | 000000+4054-4040+4112=  | 004126 |
| 4063+73=40D6 | FFFFFF6+4124-4112=      | 000008 |
| 4063+76=40D9 | FFFFFFF+4124-4112+4040= | 004051 |
| 4063+79=40DC | FFFFFF0+4054-4040=      | 000004 |
| 4063+7C=40DF | 000060+4063-4040=       | 000083 |



# Loader and Linker

Loading PROGC (**ASSIGNMENT**) :

| ADDRESS | Content |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---------|---------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
|         | 0       | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | A | B | C | D | E | F |
| 40E0    |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 40F0    |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 4100    |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 4110    |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 4120    |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 4130    |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

Updating the memory for PROGC (**ASSINGMENT**)

```
HPROGC 000000000051
DLISTC 000030ENDC 000042
RLISTA ENDA LISTB ENDB
:
T0000180C031000007710000405100000
:
T0000420F00003000000080000110000000000000
M00001905+LISTA
M00001D05+LISTB
M00002105+ENDA
M00002105-LISTA
M00004206+ENDA
M00004206-LISTA
M00004206+PROGC
M00004806+LISTA
M00004B06+ENDA
M00004B06-LISTA
M00004B06-ENDB
M00004B06+LISTB
M00004E06+LISTB
M00004E06-LISTA
E
```